

Day 2 Lectures

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This is a heading

This is the same comment that I hash out inside the code block: load the tidyverse library

```
# load the tidyverse library  
library(tidyverse)
```

```
mean(1, 2, 3)
```

```
[1] 1
```

Numeric data

```
# Create a vector of the number of pets owned by each household  
num_rats <- c(0, 1, 2, 2, 3, 1, 4, 0, 2, 1, 2, 2, 0, 3, 2, 1, 1, 4, 2, 0)  
num_rats
```

```
[1] 0 1 2 2 3 1 4 0 2 1 2 2 0 3 2 1 1 4 2 0
```

```
class(num_rats)
```

```
[1] "numeric"
```

```
num_rats_int <- as.integer(num_rats) # coercion  
num_rats_int
```

```
[1] 0 1 2 2 3 1 4 0 2 1 2 2 0 3 2 1 1 4 2 0
```

```
class(num_rats_int)
```

```
[1] "integer"
```

```
pies <- pi * seq(1:5)
pies
```

```
[1] 3.141593 6.283185 9.424778 12.566371 15.707963
```

```
class(pies)
```

```
[1] "numeric"
```

```
as.integer(pies)
```

```
[1] 3 6 9 12 15
```

```
round(pies, 2)
```

```
[1] 3.14 6.28 9.42 12.57 15.71
```

The `as.integer()` function does not care about rounding, for example:

```
as.integer(round(pies, 2))
```

```
[1] 3 6 9 12 15
```

```
set.seed(13)
# Generate a vector of 50 normally distributed temperature values
temp_data <- round(rnorm(n = 50, mean = 15, sd = 3), 2)
# temp_data <- rnorm(n = 50, mean = 15, sd = 3)
temp_data
```

```
[1] 16.66 14.16 20.33 15.56 18.43 16.25 18.69 15.71 13.90 18.32 11.72 16.39
[13] 10.92 9.43 13.68 14.42 19.19 15.30 14.66 17.11 15.79 20.51 16.07 11.86
[25] 16.86 15.45 10.62 8.92 11.83 12.82 14.98 17.54 13.85 13.42 14.18 13.18
[37] 14.00 14.28 12.41 12.46 15.30 19.77 16.70 19.84 13.59 12.82 11.93 9.19
[49] 15.83 19.23
```

```
length(temp_data)
```

```
[1] 50
```

```
mean(temp_data)
```

```
[1] 14.9212
```

```
sd(temp_data)
```

```
[1] 2.92563
```

```
class(temp_data)
```

```
[1] "numeric"
```

```
names <- c("Fluffy", "Jesse")  
names <- c('Fluffy', 'Jesse')  
class(names)
```

```
[1] "character"
```

Factor data

```
# Generate a vector of vegetation types  
  
vegetation <- sample(c("grassland", "forest", "wetland"), size = 50, replace =  
TRUE)  
  
# View the vegetation data  
vegetation
```

```
[1] "forest"    "wetland"   "grassland" "grassland" "grassland" "wetland"  
[7] "grassland" "wetland"   "grassland" "grassland" "wetland"   "forest"  
[13] "wetland"   "forest"    "forest"    "grassland" "wetland"   "grassland"  
[19] "grassland" "grassland" "forest"    "wetland"   "grassland" "forest"  
[25] "grassland" "forest"    "forest"    "grassland" "forest"    "grassland"  
[31] "grassland" "wetland"   "wetland"   "grassland" "grassland" "forest"  
[37] "wetland"   "grassland" "forest"    "forest"    "grassland" "grassland"  
[43] "forest"    "forest"    "grassland" "forest"    "grassland" "wetland"  
[49] "wetland"   "forest"
```

```
class(vegetation)
```

```
[1] "character"
```

```
veg_fact <- as.factor(vegetation)
class(veg_fact)
```

```
[1] "factor"
```

Ordered factors

```
# Vector of ordinal data representing the successional stage of a forest
coffee <- c("Small", "Tall", "Venti", "Grande", "Ginormous")
class(coffee)
```

```
[1] "character"
```

```
# Convert to ordered factor
coffee <- factor(coffee, ordered = TRUE,
                 levels = c("Small", "Tall", "Venti",
                           "Grande", "Ginormous"))
coffee
```

```
[1] Small    Tall     Venti    Grande   Ginormous
Levels: Small < Tall < Venti < Grande < Ginormous
```

```
class(coffee)
```

```
[1] "ordered" "factor"
```

Logical data

```
# Generate a vector of 1s and 0s to represent the presence
# or absence of a species in different ecological sites
species_presence <- sample(c(0, 1), 10,
                          replace = TRUE)
class(species_presence)
```

```
[1] "numeric"
```

```
species_presence_logi <- as.logical(species_presence)
class(species_presence_logi)
```

```
[1] "logical"
```

Dates

```
my_time <- as.POSIXct("2022-03-10 12:34:56")
class(my_time)
```

```
[1] "POSIXct" "POSIXt"
```

Missing values

```
# Set the length of the sequence
n <- 100

# Generate a sequence of random normal numbers with
# mean 0 and standard deviation 1
data <- rnorm(n, mean = 0, sd = 1)

# Randomly assign 5% of the values as missing
missing_indices <- sample(1:n, size = round(0.05*n))
data[missing_indices] <- NA
length(data)
```

```
[1] 100
```

```
mean(data, na.rm = TRUE)
```

```
[1] -0.1332528
```

```
length(data)
```

```
[1] 100
```

```
data_sans_na <- na.omit(data)
length(data_sans_na)
```

```
[1] 95
```

Matrices (plural) and a matrix (singular)

```
# create a numeric matrix
my_matrix_1 <- matrix(1:6, nrow = 2, ncol = 3,
                     byrow = FALSE)

my_matrix_2 <- matrix(1:6, nrow = 2, ncol = 3,
                     byrow = TRUE)
```

```
# print the matrix
my_matrix_1
```

```
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
```

```
my_matrix_2
```

```
      [,1] [,2] [,3]
[1,]    1    2    3
[2,]    4    5    6
```

```
dim(my_matrix_1)
```

```
[1] 2 3
```

```
mat <- matrix(rnorm(n = 15, mean = 0, sd = 1),
             nrow = 5, ncol = 3)
mat <- round(mat, 2)
dim(mat)
```

```
[1] 5 3
```

```
ncol(mat)
```

```
[1] 3
```

```
nrow(mat)
```

```
[1] 5
```

```
# coerce the matrix to a vector  
# 2D to 1D  
vect <- as.vector(mat)  
vect
```

```
[1] -1.04  0.82  1.00  0.43  0.47 -1.10  0.61  0.02  0.74  1.25 -0.73 -0.07  
[13]  1.03  0.82 -0.45
```

```
length(vect)
```

```
[1] 15
```

```
mat
```

```
      [,1] [,2] [,3]  
[1,] -1.04 -1.10 -0.73  
[2,]  0.82  0.61 -0.07  
[3,]  1.00  0.02  1.03  
[4,]  0.43  0.74  0.82  
[5,]  0.47  1.25 -0.45
```

```
mat[2, ]
```

```
[1]  0.82  0.61 -0.07
```

```
mat[5, c(1, 3)]
```

```
[1]  0.47 -0.45
```

Arrays

```
# create a 2-dimensional array  
my_array <- array(1:27, dim = c(3, 3, 3))
```

```
# print the array  
my_array
```

```
, , 1  
      [,1] [,2] [,3]  
[1,]    1    4    7  
[2,]    2    5    8  
[3,]    3    6    9  
  
, , 2  
      [,1] [,2] [,3]  
[1,]   10   13   16  
[2,]   11   14   17  
[3,]   12   15   18  
  
, , 3  
      [,1] [,2] [,3]  
[1,]   19   22   25  
[2,]   20   23   26  
[3,]   21   24   27
```

```
dim(my_array)
```

```
[1] 3 3 3
```

```
my_array[2, 2, 2]
```

```
[1] 14
```

```
# square bracket notation:  
# [row, column, depth]  
my_array[2, 3, 3]
```

```
[1] 26
```

```
as.vector(my_array)
```

```
[1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
25
[26] 26 27
```

Making a data.frame

```
# Create a vector of dates
dates <- as.Date(c("2022-01-01", "2022-01-02", "2022-01-03",
                  "2022-01-04", "2022-01-05"))

# Create a vector of numeric data
numeric_data <- rnorm(n = 5, mean = 0, sd = 1)

# Create a vector of categorical data
categorical_data <- c("A", "B", "C", "A", "B")

# Combine the vectors into a data.frame
my_dataframe <- data.frame(day = dates,
                           number = numeric_data,
                           category = categorical_data)

# Print the dataframe
my_dataframe
```

	day	number	category
1	2022-01-01	1.1059215	A
2	2022-01-02	1.2593616	B
3	2022-01-03	-0.6061951	C
4	2022-01-04	-1.4083780	A
5	2022-01-05	1.6957718	B

```
class(my_dataframe)
```

```
[1] "data.frame"
```

```
colnames(my_dataframe)
```

```
[1] "day"      "number"   "category"
```

```
rownames(my_dataframe)
```

```
[1] "1" "2" "3" "4" "5"
```

```
dim(my_dataframe)
```

```
[1] 5 3
```

```
str(my_dataframe)
```

```
'data.frame':  5 obs. of  3 variables:
 $ day      : Date, format: "2022-01-01" "2022-01-02" ...
 $ number   : num  1.106 1.259 -0.606 -1.408 1.696
 $ category : chr  "A" "B" "C" "A" ...
```

```
mean(my_dataframe$number)
```

```
[1] 0.4092964
```

```
range(my_dataframe$day)
```

```
[1] "2022-01-01" "2022-01-05"
```

```
summary(my_dataframe)
```

	day	number	category
Min.	:2022-01-01	Min. : -1.4084	Length:5
1st Qu.	:2022-01-02	1st Qu.: -0.6062	Class :character
Median	:2022-01-03	Median : 1.1059	Mode :character
Mean	:2022-01-03	Mean : 0.4093	
3rd Qu.	:2022-01-04	3rd Qu.: 1.2594	
Max.	:2022-01-05	Max. : 1.6958	

```
glimpse(my_dataframe)
```

```
Rows: 5
Columns: 3
$ day      <date> 2022-01-01, 2022-01-02, 2022-01-03, 2022-01-04, 2022-01-05
```

```
$ number    <dbl> 1.1059215, 1.2593616, -0.6061951, -1.4083780, 1.6957718
$ category  <chr> "A", "B", "C", "A", "B"
```

```
my_tibble <- as_tibble(my_dataframe)
class(my_tibble)
```

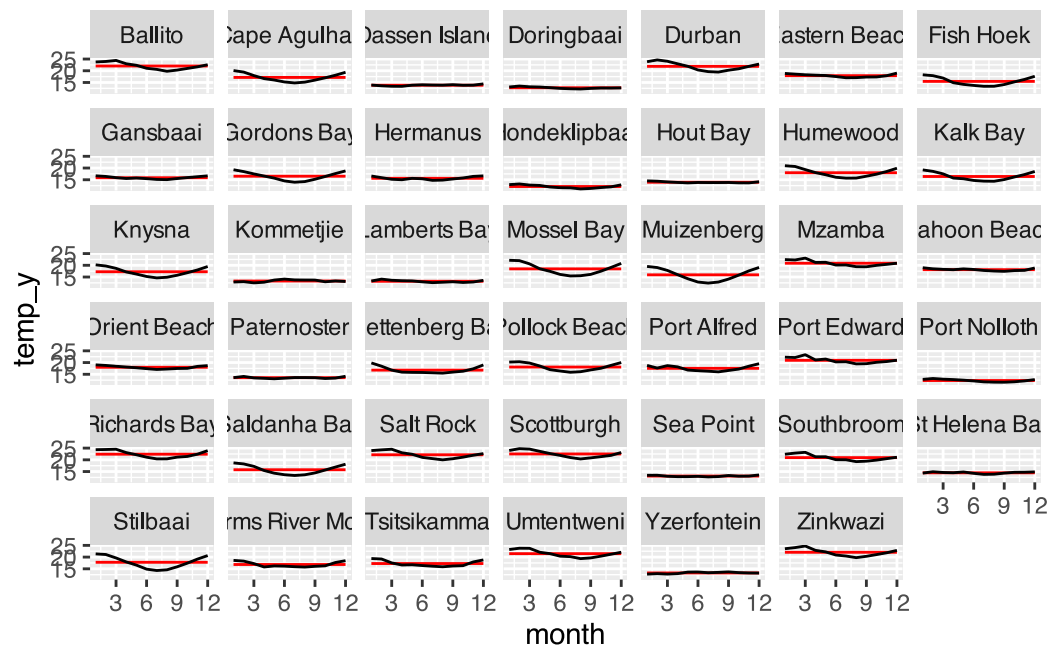
```
[1] "tbl_df"      "tbl"        "data.frame"
```

```
my_tibble
```

```
# A tibble: 5 × 3
  day          number category
<date>      <dbl> <chr>
1 2022-01-01   1.11    A
2 2022-01-02   1.26    B
3 2022-01-03  -0.606   C
4 2022-01-04  -1.41    A
5 2022-01-05   1.70    B
```

Exercise in frustration

```
read_csv("../data/SAMOS/other_data/SACTN_SAWS.csv") |>
  mutate(month = as.integer(month(date, label = FALSE))) |>
  group_by(site) |>
  mutate(temp_y = mean(temp, na.rm = TRUE)) |>
  group_by(site, month) |>
  summarise(temp_m = mean(temp, na.rm = TRUE),
            temp_y = mean(temp_y)) |>
  ggplot(aes(x = month)) +
    geom_line(aes(y = temp_y), colour = "red") +
    geom_line(aes(y = temp_m)) +
    scale_x_continuous(breaks = c(3, 6, 9, 12)) +
    facet_wrap(~site)
```



Reading in data into R

For this exercise, we will use the `laminaria.csv` file. It is located in the `data/SAMOS/other_data` folder.

Bibliography